4.1 Complex Numbers

Pre-Calculus 4.1 · Student sheet

Name _____ Period ____ Date _____

I can…

- write square roots of negative numbers using i.
- add, subtract, and multiply complex numbers.
- divide complex numbers using the complex conjugate.

Vocabulary · fill in the missing word

Imaginary unit i — i = _____, so $i^2 = -1$

Complex number — a number a + bi, where a is the real part and b is the _____ part

Complex conjugates — a _____ of complex numbers a + bi and a − bi

Principal square root of a negative number — $\sqrt{-a}$ = _____, for a > 0

Part 1 · Powers of i

Use $i^2 = -1$. Each power is the one before times i.

i^1	i^2	i^3	i^4	i^5	i^6	i^7	i^8
i	−1						

i^{23} = _____ i^{100} = _____

Part 2 · Notes and examples (fill in each note, then work the example)

The imaginary unit: i = $\sqrt{-1}$, so i^2 = _____.

Example 1

Find a and b: (a − 1) + (b + 3)i = 5 + 8i

Adding and subtracting: combine the _____ parts and the imaginary parts.

Example 2

a. (3 − i) + (2 + 3i)

b. (4 + 2i) − (7 − 5i)

Multiplying: use FOIL, then replace i^2 with _____.

Example 3

a. (2 − i)(4 + 3i)

b. (3 + 2i)(3 − 2i)

Dividing: multiply the numerator and denominator by the _____ of the denominator.

Example 4

Write in standard form: $\frac{2 + 3i}{4 - 2i}$

Square roots of negatives: write $\sqrt{-a}$ as _____ BEFORE multiplying.

Example 5

a. Simplify $\sqrt{-3} \cdot \sqrt{-12}$.

b. Solve $3x^2 - 2x + 5 = 0$.

Part 3 · Partner practice

Partner 1

Multiply: $(5 + 2i)(3 - 4i)$

Partner 2

Write in standard form: $\frac{3 + i}{1 - 2i}$